

Lucas Numbers

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1 Introduction

The Lucas sequence is an intriguing and relatively mysterious series of numbers named after François Édouard Anatole Lucas (1842–91) who studied this sequence in addition to the closely related Fibonacci sequence.

Lucas numbers can be constructed by starting from 2, then 1, and defining the following members of the sequence as the sum of the two prior. This rule is similar to the rule used for Fibonacci sequence which can be thought of starting from 0 and 1 and then any following member is received as the sum of the prior two.

Lucas sequence members are received by taking the prior and following-number Fibonacci sequence member. It can also be thought of as the same-number Fibonacci member plus twice the prior Fibonacci number, see Figure 1.

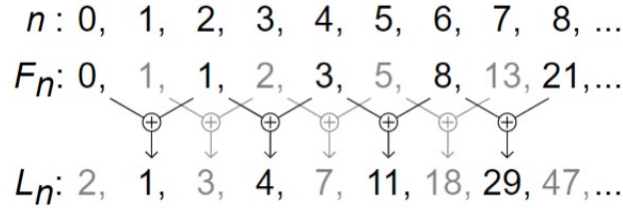


Figure 1: Lucas' relation to the Fibonacci numbers, with $L(n) = F(n-1) + F(n+1)$, or $L(n) = F(n) + 2F(n-1)$. It can be shown that $L(n) = \frac{F(2n)}{F(n)}$

2 Relation to the Golden Ratio

While having complex symmetries, Lucas numbers can be shown as related to the golden ratio and (1-golden ratio ϕ),

$$\phi = \frac{1+\sqrt{5}}{2}, \text{ and}$$

$$(1 - \phi) = \frac{2 - 1 - \sqrt{5}}{2} = \frac{1 - \sqrt{5}}{2} = \frac{(1 - \sqrt{5})(1 + \sqrt{5})}{2(1 + \sqrt{5})} = -\frac{2}{1 + \sqrt{5}} = -\frac{1}{\phi}$$

Denoting $\hat{\phi} = -\frac{1}{\phi}$, we can prove that Lucas sequence members equal to

$$L(n) = \phi^n + (1 - \phi)^n = \left(\frac{1 + \sqrt{5}}{2}\right)^n + \left(\frac{1 - \sqrt{5}}{2}\right)^n = \phi^n + \hat{\phi}^n$$

This proof can be done by mathematical induction.

The basis of induction is $L(0)=1+1=2$ which is true, and $L(1) = \left(\frac{1+\sqrt{5}}{2}\right)^1 + \left(\frac{1-\sqrt{5}}{2}\right)^1 = 1$ which is also true.

As the second part of the proof, induction step, we assume that for any j such that $0 < j \leq k+1$, $L(j) = \phi^j + \hat{\phi}^j$, and we will show that $L(k+2) = \phi^{k+2} + \hat{\phi}^{k+2}$.

We will use an interesting property of the golden ratio, that $\phi^2 = 1 + \phi$, and $\hat{\phi}^2 = 1 + \hat{\phi}$.

In other words, ϕ and $\hat{\phi}$ are roots of the equation $x^2 = 1 + x$.

This is because

$$\phi^2 = \left(\frac{1+\sqrt{5}}{2}\right)^2 = \frac{(1+\sqrt{5})^2}{4} = \frac{6+2\sqrt{5}}{4} = \frac{3+\sqrt{5}}{2} = 1 + \frac{1+\sqrt{5}}{2} = 1 + \phi$$

and

$$\hat{\phi}^2 = \left(\frac{1-\sqrt{5}}{2}\right)^2 = \frac{(1-\sqrt{5})^2}{4} = \frac{6-2\sqrt{5}}{4} = \frac{3-\sqrt{5}}{2} = 1 + \frac{1-\sqrt{5}}{2} = 1 + \hat{\phi}.$$

Then

$$\phi^{k+2} + \hat{\phi}^{k+2} = (1 + \phi)\phi^k + (1 + \hat{\phi})\hat{\phi}^k = (\phi^k + \hat{\phi}^k) + (\phi^{k+1} + \hat{\phi}^{k+1}) = L(k) + L(k+1) = L(k+2)$$

which completes the proof by mathematical induction.

3 Cyclicity

One more interesting property of Lucas numbers is their cyclicity using certain mods (remainders), see Figure 2. For example, for (mod 2) cyclicity, let's apply (mod 2) to the below equality:

$$\begin{aligned} L(n) - L(n-3) &\equiv L(n-1) + L(n-2) - L(n-3) \\ &\equiv 2L(n-2) + L(n-3) - L(n-3) \\ &\equiv 2L(n-2) \end{aligned}$$

We get $L(n) \equiv L(n-3) \pmod{2}$ meaning repetition every 3 numbers.

For (mod 4) cyclicity, let's apply (mod 4) to below equality, which uses above:

$$\begin{aligned} L(n) - L(n-6) &\equiv 2L(n-2) + L(n-3) - L(n-6) \equiv 2L(n-3) + 2L(n-4) + L(n-3) - L(n-6) \\ &\equiv 3L(n-3) + 2L(n-4) - L(n-6) \\ &\equiv 3L(n-3) + L(n-4) + L(n-5) + L(n-6) - L(n-6) \\ &\equiv 3L(n-3) + L(n-3) = 4L(n-3) \end{aligned}$$

$L(n) \equiv L(n-6) \pmod{4}$, meaning repetition every 6 numbers.

For (mod 5) cyclicity, let's apply (mod 5) to below equality, which uses the ideas above plus Fibonacci representation $L(n) = F(n-1) + F(n+1)$:

$$\begin{aligned}
L(n)-L(n-4) &\equiv 2L(n-2) + L(n-3)-L(n-4) \\
&\equiv 3L(n-3) + 2L(n-4) - L(n-4) = 3L(n-3) + L(n-4) \\
&\equiv 3F(n-2) + 3F(n-4) + F(n-3) + F(n-5) \\
&\equiv 3F(n-2) + 2F(n-3) + 2F(n-4) \\
&\equiv 5F(n-2)
\end{aligned}$$

$L(n) \equiv L(n-4) \pmod{5}$, meaning repetition every 4 numbers. Figure 2 also illustrates repetition every 12 numbers by (mod 40) and every 60 numbers by (mod 100).

	L(1)	(Mod 2)	(Mod 4)	(Mod 5)	(Mod 40)	(Mod 100)
1	2	0	2	2	2	2
2	1	1	1	1	1	1
3	3	1	3	3	3	3
4	4	0	0	4	4	4
5	7	1	3	2	7	7
6	11	1	3	1	11	11
7	18	0	2	3	18	18
8	29	1	1	4	29	29
9	47	1	3	2	7	47
10	76	0	0	1	36	76
11	123	1	3	3	3	23
12	199	1	3	4	39	99
13	322	0	2	2	2	22
14	521	1	1	1	1	21
15	843	1	3	3	3	43
16	1,364	0	0	4	4	64
17	2,207	1	3	2	7	7
18	3,571	1	3	1	11	71
19	5,778	0	2	3	18	78
20	9,349	1	1	4	29	49
21	15,127	1	3	2	7	27
22	24,476	0	0	1	36	76
23	39,603	1	3	3	3	3
24	64,079	1	3	4	39	79
25	103,682	0	2	2	2	82
26	167,761	1	1	1	1	61
27	271,443	1	3	3	3	43
28	439,204	0	0	4	4	4
29	710,647	1	3	2	7	47
30	1,149,851	1	3	1	11	51
31	1,860,498	0	2	3	18	98
32	3,010,349	1	1	4	29	49
33	4,870,847	1	3	2	7	47
34	7,881,196	0	0	1	36	96
35	12,752,043	1	3	3	3	43
36	20,633,239	1	3	4	39	39
37	33,385,282	0	2	2	2	82
59	1,322,157,322,203	1	3	3	3	3
60	2,139,295,485,799	1	3	4	39	99
61	3,461,452,808,002	0	2	2	2	2
62	5,600,748,293,801	1	1	1	1	1

Figure 2.

Periodic properties of the Lucas numbers:

- cycles length 3 using (mod 2) colored yellow,

- cycles length 6 using (mod 4) colored green,
- cycles length 4 using (mod 5) colored blue, and
- cycles length 60 using (mod 100) — purple highlights that (mod 100), $L(61) \equiv L(1)$ and $L(62) \equiv L(2)$.

4 Lucas, Fibonacci numbers and Pascal's triangle

Figure 3 illustrates a known representation of Fibonacci numbers using Pascal's triangle discovered by Francois Lucas, algebraically written as

$$F(n) = \sum_{k=0}^{\lfloor (n-1)/2 \rfloor} \binom{n-k-1}{k}$$

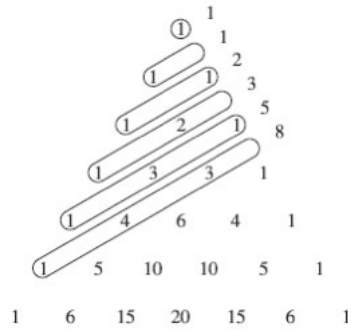
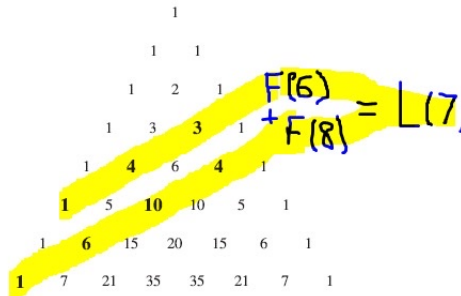


Figure 3: Fibonacci numbers in the Pascal's triangle.

Using the Lucas sequence representation as a sum of prior and the next Fibonacci $L(n) = F(n-1) + F(n+1)$, Pascal's triangle alternate diagonals can be used to receive Lucas numbers, see Figure 4. Algebraically,

$$F(n) = \sum_{k=0}^{\lfloor n/2 \rfloor} \left[\binom{n-k}{k} + \binom{n-k-1}{k-1} \right]$$



Figuer 4: Lucas numbers are sums of alternate diagonals of Pascal's triangle.

This leads to a rewrite of the formula in a combinatorics-like manner.

$$F(n) = \sum_{k=0}^{\lfloor n/2 \rfloor} \left(\frac{n-k}{k} + \frac{n-k-1}{k-1} \right) = \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{n}{n-k} \left(\frac{n-k}{k} \right)$$

5 Notes on Applications

Lucas sequence applications include identifying Mersenne primes (these have form $2^n - 1$), primality tests, solving congruences, integer factorization, solving Diophantine equations and cryptography. As such, Lucas sequences' practical applications are in computational number theory.

The Lucas-Lehmer test for numbers $M_n = 2^n - 1$ was used to identify all known Mersenne primes for $n > 127$.

After applying the theory of his sequences to Mersenne primes, Lucas applied it to primality testing of numbers of the form $Ap^n \pm 1$ where p is prime.

A number of problems concerning Lucas sequences or their possible generalizations remain unsolved or unexplored.

In 1993, a public-key cryptosystem, LUC was invented based on Lucas sequences theory. It was constructed as a viable alternative to RSA encryption. While LUC is somewhat more complex, it has been recommended as a possible authentication system and continues to be the object of active research.

6 Conclusion

It is remarkable that after a lot of activity devoted to a simple pair of sequences, Fibonacci and Lucas, there are still a number of unanswered questions acutely important to today's mathematicians as well as to computer scientists and electrical engineers awaiting solutions. The sequence OEIS number is A000032 and it is listed at <https://oeis.org/A000032>.

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